

YANKTON GURNEY, SD, USA

WMO: 726525

Lat: 42.917N

Lon: 97.383W

Elev: 1172

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-2.3	-17.7	2.2	-6.2	-10.9	3.2	-2.6	36.5	12.2	33.3	18.0	9.2	310	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	92.9	74.1	90.2	73.5	87.7	72.5	77.7	88.1	75.8	85.7	74.2	83.9	14.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	135.3	84.4	72.7	126.5	82.4	71.6	122.1	81.1	42.0	87.7	40.1	86.0	38.5	83.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.8	26.1	23.2	-14.9	98.0	5.7	3.5	-19.0	100.5	-22.3	102.6	-25.6	104.5	-29.7	107.0		
			-15.5	80.8	5.1	1.8	-19.2	82.1	-22.2	83.1	-25.0	84.2	-28.7	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	20.7	24.4	36.1	48.0	59.7	70.0	75.0	72.0	64.5	50.7	36.8	24.3
	DBStd	21.49	12.91	12.90	12.89	10.18	8.62	7.06	6.17	5.57	8.40	9.14	11.12	11.68
	HDD50	3571	909	718	457	152	16	0	0	4	103	414	798	
	HDD65	6848	1374	1138	896	516	211	32	4	9	110	449	846	1263
	CDD50	3076	0	1	27	92	317	599	775	683	439	126	17	0
	CDD65	878	0	0	1	6	47	181	314	227	95	7	0	0
	CDH74	8684	0	0	18	109	535	1797	3256	1937	934	94	4	0
	CDH80	3207	0	0	2	29	166	653	1362	665	307	23	0	0
Wind	WSAvg	10.4	11.0	11.2	11.7	12.3	11.0	9.7	8.6	8.1	9.5	10.4	10.4	10.6
Precipitation	PrecAvg	23.2	0.4	0.5	1.7	2.2	3.5	3.8	3.1	2.9	2.4	1.5	1.0	0.6
	PrecMax	35.4	1.3	1.8	5.2	7.1	11.7	7.0	8.9	7.8	7.4	5.5	4.2	2.0
	PrecMin	14.6	0.0	0.0	0.0	0.4	0.7	0.9	0.2	0.3	0.7	0.1	0.0	0.0
	PrecStd	5.2	0.4	0.4	1.3	1.4	2.4	1.7	2.1	1.9	1.6	1.5	1.0	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	62.7	75.4	83.9	90.4	95.2	99.6	95.4	91.2	82.2	72.4	55.2
		MCWB	43.3	49.3	55.7	62.7	67.6	74.2	74.8	74.7	71.2	62.1	54.6	44.3
	2%	DB	46.3	53.9	68.5	75.8	84.0	90.5	93.4	90.4	87.9	75.1	64.3	48.4
		MCWB	38.4	44.7	54.2	57.9	65.4	72.5	75.3	74.7	70.8	60.0	52.0	40.9
	5%	DB	42.5	47.6	61.3	70.4	80.7	87.6	90.6	87.6	83.3	71.7	58.8	44.6
		MCWB	36.1	40.6	50.2	55.8	63.9	70.4	75.3	73.7	68.7	58.2	47.7	37.8
	10%	DB	37.2	42.6	54.8	64.4	75.1	83.6	87.8	83.0	79.5	66.0	53.8	39.0
		MCWB	33.2	36.5	46.0	52.0	61.3	69.0	74.3	71.7	65.7	55.2	45.3	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	50.8	60.3	64.4	71.4	77.8	80.5	79.2	75.4	66.3	58.0	46.8
		MCDB	54.6	59.6	68.9	80.1	83.6	90.0	90.7	89.2	84.8	77.4	66.9	52.8
	2%	WB	39.2	44.7	55.6	60.2	68.2	74.7	78.3	76.8	73.0	62.8	53.0	41.2
		MCDB	45.8	53.3	66.1	73.2	80.3	86.0	88.6	86.4	82.9	71.5	63.3	48.6
	5%	WB	36.4	40.3	51.2	56.7	65.6	72.7	76.8	75.1	70.8	59.6	48.9	37.8
		MCDB	41.5	47.1	62.4	68.3	77.3	84.0	86.9	84.0	80.7	68.2	58.4	43.4
	10%	WB	33.6	36.6	46.3	53.5	62.9	70.6	75.0	73.3	68.3	56.3	45.1	34.9
		MCDB	37.5	41.9	54.6	63.3	73.2	81.4	84.9	81.2	77.0	65.4	52.5	39.0
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.3	19.6	21.0	20.4	20.0	20.0	19.8	21.9	21.1	19.5	16.3
		MCDBR	24.1	26.8	29.8	30.3	26.7	24.3	23.1	23.9	25.9	28.9	28.8	23.6
		MCWBR	17.4	18.3	18.3	16.4	13.3	11.4	10.5	11.8	12.8	16.3	17.9	16.8
	5% WB	MCDBR	22.8	25.6	28.7	26.7	23.6	21.4	20.1	20.4	22.2	23.7	27.8	22.3
		MCWBR	17.0	17.8	17.7	15.4	13.2	11.7	11.2	11.7	12.4	14.9	18.0	16.2
Clear-Sky Solar Irradiance	taub	0.275	0.291	0.319	0.376	0.394	0.404	0.409	0.411	0.360	0.324	0.289	0.275	
	taud	2.447	2.432	2.423	2.272	2.305	2.331	2.322	2.304	2.415	2.491	2.504	2.461	
	Ebn at Noon	277	291	294	282	279	276	273	268	276	274	270	266	
	Edh at Noon	23	28	32	41	41	40	40	39	32	26	22	21	
All-Sky Solar Radiation	RadAvg	617	937	1230	1561	1825	2047	2124	1773	1423	979	678	504	
	RadStd	50	67	100	120	134	143	91	133	106	114	52	44	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page